

Lab 19 Worksheet — Hypervisor Types and Virtual Machine Resource Planning

Name: ____ Date: ____

Exam objective: Distinguish Type 1 from Type 2 hypervisors, plan virtual machine resource allocation and identify the firmware prerequisite for virtualization (Core 1 objective 4.2).

Record as you go

Step	What you did	What you observed
1	Build the comparison table with rows for Type 1 and Type 2 and columns for where it runs, examples, typical use, performance and management overhead.	
2	Record that a Type 1 bare-metal hypervisor such as VMware ESXi, Microsoft Hyper-V Server or Citrix Hypervisor runs directly on hardware with no host OS beneath it.	
3	Record that a Type 2 hosted hypervisor such as VirtualBox, VMware Workstation or Parallels runs as an application on top of a desktop operating system.	
4	Explain why Type 1 performs better — there is no host operating system competing for CPU and memory or adding a scheduling layer.	
5	Check whether hardware virtualization support is enabled on your host, since no 64-bit guest will start without it.	
6	Confirm the same from Task Manager by opening Performance then CPU and reading the Virtualization field.	
7	If virtualization is disabled, record the exact BIOS/UEFI menu path where Intel VT-x or AMD-V is enabled, and note that this is the single	

Step	What you did	What you observed
	most common cause of a VM refusing to start.	
8	Record the host's total physical resources: CPU cores and logical processors, total RAM, and free disk space.	
9	Plan four VMs — a Windows desktop, a Linux server, a Windows server and a test machine — assigning vCPU, RAM and disk to each based on the guest OS minimum plus a working margin.	
10	Sum the planned RAM across all four VMs and confirm the total leaves at least 4 GB for the host, since over-committing RAM is the fastest way to bring a host to a halt.	
11	Explain the difference between a thick-provisioned disk, which claims all its space immediately, and a thin-provisioned disk, which grows as it fills, and state the risk of over-committing thin disks.	
12	Explain what a snapshot does, why it is not a backup, and why leaving snapshots in place for weeks degrades performance and consumes disk.	

Verification

Success criterion: Virtualization support is confirmed enabled on the host, your allocation plan's total RAM leaves at least 4 GB for the host, and every VM is allocated at or above its guest OS minimum with the margin stated.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
